

Backend Development – Phase 2

ServiceNow Project Documentation

Purpose: This document provides a GitHub-ready overview of the Backend Development Phase 2 project implemented on ServiceNow.

1. Project Overview

The Backend Development Phase 2 project focuses on building server-side functionality in ServiceNow. The project demonstrates how business requirements can be implemented using tables, server-side scripts, Business Rules, Script Includes, data processing, and controlled automation.

2. Objectives

- Create and manage application data using ServiceNow tables.
- Implement server-side business logic.
- Automate data validation and processing.
- Use reusable server-side scripting components.
- Understand how backend components interact with ServiceNow records.
- Test and document the implemented functionality for deployment and GitHub submission.

3. Technologies and Platform

Component	Technology
Platform	ServiceNow
Backend	ServiceNow server-side scripting
Scripting	JavaScript
Data	ServiceNow tables and GlideRecord
Automation	Business Rules / server-side logic
Version Documentation	GitHub

4. Backend Development Components

Tables

Store project records and provide the data structure required by the application.

Business Rules

Execute server-side logic automatically when records are inserted, updated, deleted, or queried, depending on configuration.

Script Includes

Provide reusable server-side JavaScript functions that can be called by other backend components.

GlideRecord

Used to query, create, update, and delete records in ServiceNow tables.

Validation

Checks data before processing so that incorrect or incomplete records can be handled appropriately.

5. Implementation Flow

- User or system creates/updates a record.
- ServiceNow receives the transaction.
- Configured Business Rules are evaluated.
- Server-side scripts validate or process the data.
- GlideRecord is used when database records must be queried or modified.
- The final record state is stored in the ServiceNow instance.
- The result is verified through testing and logs.

6. Sample Server-Side Script Pattern

```
var gr = new GlideRecord('YOUR_TABLE');
gr.addQuery('active', true);
gr.query();

while (gr.next()) {
  gs.info('Record: ' + gr.getDisplayValue());
}
```

Replace YOUR_TABLE with the table used in the actual project. This example demonstrates the general GlideRecord query pattern.

7. Testing

- Verify that records can be created and updated correctly.
- Check Business Rule execution under the configured conditions.
- Validate mandatory fields and business conditions.
- Confirm that server-side scripts produce the expected result.
- Review system logs for errors or unexpected behavior.
- Test both valid and invalid input scenarios.

8. Expected Outcome

The completed Phase 2 implementation should provide reliable backend processing within ServiceNow, reduce manual operations through automation, maintain consistent data, and demonstrate practical use of ServiceNow server-side development concepts.

9. Suggested GitHub Repository Structure

```
ServiceNow-Backend-Development-Phase2/
■■■ README.md
■■■ documentation/
■ ■■■ ServiceNow_Backend_Development_Phase_2_Project.pdf
■■■ scripts/
■ ■■■ server-side-scripts.txt
■■■ screenshots/
```

- ■■■ business-rule.png
- ■■■ script-include.png
- ■■■ testing.png
- notes/
- implementation-notes.md

10. Conclusion

Backend Development Phase 2 demonstrates the implementation of server-side functionality in ServiceNow. The project combines structured data, backend scripting, automation, validation, and testing to create a maintainable ServiceNow application. The documentation can be included in a GitHub repository along with project screenshots and source/configuration notes.

Note: Add your actual project name, table names, Business Rules, Script Includes, screenshots, and test results before final submission.